Background: Results Used from Chapters 1–3

How to use this

Chapters 4–7 rely on standard facts about Lebesgue measure, the Lebesgue integral and L^p spaces that were developed in Chapters 1–3. They are collected here, in the order in which they arise, and are cited in the later chapters as *Fact B.1*, *Fact B.2*, ... Throughout, $(X, \mathcal{S}, \mu)$ is a measure space, m is Lebesgue measure on $\mathbb{R}$ with σ-algebra $\mathcal{M}$ of Lebesgue measurable sets, and $1 \le p \le \infty$.

Measures

Fact B.1 (translation invariance)

For $E \in \mathcal{M}$ and $x \in \mathbb{R}$, $E + x \in \mathcal{M}$ and $m(E + x) = m(E)$; consequently $\int f(y + x)\, dm(y) = \int f(y)\, dm(y)$ for integrable f.

Fact B.2 (completeness of Lebesgue measure)

$(\mathbb{R}, \mathcal{M}, m)$ is complete: every subset of a set of Lebesgue measure zero is Lebesgue measurable (and has measure zero).

Fact B.3 (regularity of Lebesgue measurable sets)

For every $S \in \mathcal{M}$ there are an F_σ-set A and a G_δ-set B with $A \subset S \subset B$ and $m(B \setminus A) = 0$.

Fact B.4 (counting measure)

On any set X with $\mathcal{S} = 2^X$, the counting measure $c(E) = \#E$ (the number of elements, possibly ∞) is a measure. On $X = \mathbb{N}$, $\int f\, dc = \sum_n f(n)$ for $f \ge 0$.

Measurable functions and the integral

Fact B.5 (approximation by simple functions)

If $f : X \to [0, \infty]$ is measurable, there is an increasing sequence of measurable simple functions $0 \le s_1 \le s_2 \le \cdots$ with $s_n \to f$ pointwise.

Fact B.6 (series of characteristic functions)

Every measurable $f : X \to [0, \infty]$ can be written as $f = \sum_{k=1}^{\infty} c_k \chi_{E_k}$ with constants $c_k \ge 0$ and measurable sets E_k.

Fact B.7 (measures defined by densities)

If $h : X \to [0, \infty]$ is measurable, then $\lambda(E) = \int_E h\, d\mu$ $(E \in \mathcal{S})$ defines a measure on $\mathcal{S}$.

Fact B.8 (functions with vanishing integrals)

(a) If $h \geq 0$ and $\int_X h\,d\mu = 0$ then $h = 0$ almost everywhere. (b) If h is integrable and $\int_E h\,d\mu = 0$ for every $E \in \mathcal{S}$, then $h = 0$ almost everywhere.

Fact B.9 (triangle inequality for integrals)

For integrable f, $\left|\int_X f\,d\mu\right| \leq \int_X |f|\,d\mu$.

Fact B.10 (linearity of the integral)

If f, g are integrable and $a, b \in \mathbb{C}$, then $af + bg$ is integrable and $\int (af + bg)\,d\mu = a\int f\,d\mu + b\int g\,d\mu$.

Fact B.11 (differentiation under the integral sign)

Let $f(x,t)$ be integrable in x for each t in an open interval I, differentiable in t, and suppose $|\partial_t f(x,t)| \leq g(x)$ for an integrable g independent of t. Then $F(t) = \int f(x,t)\,d\mu(x)$ is differentiable on I with $F'(t) = \int \partial_t f(x,t)\,d\mu(x)$.

Fact B.12 (periodisation)

For $f \in L^1(\mathbb{R})$ the series $F(t) = \sum_{n\in\mathbb{Z}} f(t - 2n\pi)$ converges absolutely for almost every t, is 2π-periodic, and $\int_{-\pi}^{\pi} \sum_n |f(t-2n\pi)|\,dt = \int_{\mathbb{R}} |f|\,dm$ (term-by-term integration, Fact B.25).

Fact B.13 (Riemann integral via step functions)

For bounded $f : [a,b] \to \mathbb{R}$ and a partition $P = \{a = x_0 < x_1 < \cdots < x_n = b\}$ put $M_i = \sup_{[x_{i-1},x_i]} f$, $m_i = \inf_{[x_{i-1},x_i]} f$ and define the step functions

$$U_P = \sum_{i=1}^{n} M_i \chi_{(x_{i-1},x_i]}, \qquad L_P = \sum_{i=1}^{n} m_i \chi_{(x_{i-1},x_i]}.$$

f is Riemann integrable iff $\inf_P \int U_P = \sup_P \int L_P$, and then this common value is $\int_a^b f$. Improper Riemann integrals are defined as limits of such integrals; Lebesgue integrals need no such limiting procedure.

Differentiation

Fact B.14 (derivative of a monotone function)

If $\alpha : [a,b] \to \mathbb{R}$ is nondecreasing, then α' exists almost everywhere, $\alpha' \geq 0$ is integrable, and $\int_{[a,b]} \alpha'\,dm \leq \alpha(b) - \alpha(a)$.

Fact B.15 (absolute continuity of the integral)

If $f \in L^1(\mathbb{R})$, then for every $\epsilon > 0$ there is $\delta > 0$ such that $m(E) < \delta$ implies $\int_E |f|\,dm < \epsilon$.

Fact B.16 (fundamental theorem for absolutely continuous functions)

If $\alpha : [a, b] \to \mathbb{R}$ is absolutely continuous, then α' exists almost everywhere, is integrable, and $\alpha(x) - \alpha(a) = \int_{[a,x]} \alpha' \, dm$ for all $x \in [a, b]$.

L^p spaces

Fact B.17 (completeness of L^p)

For $1 \le p \le \infty$, $L^p(\mu)$ is a Banach space.

Fact B.18 (a.e. convergent subsequence)

If $\|f_n - f\|_p \to 0$ for some $1 \le p < \infty$, then a subsequence $f_{n_k} \to f$ almost everywhere.

Fact B.19 (translates)

For a function f on $\mathbb{R}$ and $s \in \mathbb{R}$, the translate is $f_s(x) = f(x - s)$.

Fact B.20 (density of simple functions)

For $1 \le p < \infty$, the measurable simple functions s with $\mu(\{s \neq 0\}) < \infty$ are dense in $L^p(\mu)$; moreover every $f \in L^p(\mu)$ is the L^p-limit of simple functions s_n with $|s_n| \le |f|$.

Fact B.21 (density of continuous functions on an interval)

For $1 \le p < \infty$, the continuous functions on $[a, b]$ are dense in $L^p[a, b]$.

Fact B.22 (density of $C_c(\mathbb{R})$)

For $1 \le p < \infty$, the continuous functions with compact support are dense in $L^p(\mathbb{R})$.

Fact B.23 (continuity of translation)

For $f \in L^p(\mathbb{R})$ and $1 \le p < \infty$, $\|f_s - f\|_p \to 0$ as $s \to 0$ (with f_s as in Fact B.19).

Further properties of measures, functions and sums

Fact B.24 (continuity of measures)

Let $E_1, E_2, \dots \in \mathcal{S}$. (a) Countable additivity: $\mu(\bigsqcup_n E_n) = \sum_n \mu(E_n)$ for disjoint E_n. (b) Continuity from below: if $E_1 \subset E_2 \subset \cdots$, then $\mu(\bigcup_n E_n) = \lim_n \mu(E_n)$. (c) Continuity from above: if $E_1 \supset E_2 \supset \cdots$ and $\mu(E_1) < \infty$, then $\mu(\bigcap_n E_n) = \lim_n \mu(E_n)$.

Fact B.25 (term-by-term integration)

For measurable $f_n : X \to [0,\infty]$, $\int_X \sum_{n=1}^\infty f_n \, d\mu = \sum_{n=1}^\infty \int_X f_n \, d\mu$ (a consequence of the monotone convergence theorem).

Fact B.26 (operations preserving measurability)

Let f, g be measurable (real or complex valued, with respect to a fixed σ-algebra). (a) $f+g$ and fg are measurable. (b) Pointwise sup, inf, lim sup, lim inf and limits of sequences of measurable functions are measurable; in particular a pointwise convergent series of measurable functions is measurable. (c) $|f|$, $\operatorname{Re} f$, $\operatorname{Im} f$ are measurable. (d) If $g : \mathbb{R} \to \mathbb{R}$ is continuous, then $g \circ f$ is measurable.

Fact B.27 (averages in a closed set)

Let $\mu(X) < \infty$, $g \in L^1(\mu)$ and S a closed subset of $\mathbb{C}$. If $\frac{1}{\mu(E)} \int_E g \, d\mu \in S$ for every $E \in \mathcal{S}$ with $\mu(E) > 0$, then $g(x) \in S$ for almost every x. (For σ-finite μ, apply this on each set of finite measure.)

Fact B.28 (sums over arbitrary index sets)

For $a_\alpha \geq 0$ ($\alpha \in A$, A any set), $\sum_{\alpha \in A} a_\alpha := \sup\{\sum_{\alpha \in F} a_\alpha : F \subset A \text{ finite}\}$. Hence if $\sum_{\alpha \in F} a_\alpha \leq M$ for every finite $F \subset A$, then $\sum_{\alpha \in A} a_\alpha \leq M$; and if the sum is finite, $a_\alpha \neq 0$ for at most countably many α.

Fact B.29 (inclusions between L^p spaces)

If $\mu(X) < \infty$ and $1 \leq p < q \leq \infty$, then $L^q(\mu) \subset L^p(\mu)$ and $\|f\|_p \leq \mu(X)^{1/p-1/q} \|f\|_q$. In particular, on $\mathbb{T}$ with total mass 1, $L^2(\mathbb{T}) \subset L^1(\mathbb{T})$ and $\|f\|_1 \leq \|f\|_2$.